SOLUTIONS - Exercise - Week 10: MATH1061/7861

(1) Define a relation ρ on $\mathbb{Z}_{\geq 1}$ by $a\rho b$ if
$$\gcd(a, 144) = \gcd(b, 144).$$

(a) **(3 marks)** Prove that ρ is an equivalence relation.

Reflexive: For any $a \in \mathbb{Z}_{\geq 1}$, $\gcd(a, 144) = \gcd(a, 144)$, and so $a\rho a$.

Symmetric: For any $a, b \in \mathbb{Z}_{\geq 1}$, suppose that $a\rho b$. This means that $\gcd(a, 144) = \gcd(b, 144)$, so $\gcd(b, 144) = \gcd(a, 144)$, which implies that $b\rho a$.

Transitive: For any $a, b, c \in \mathbb{Z}_{\geq 1}$, suppose that $a\rho b$ and $b\rho c$. This means that $\gcd(a, 144) = \gcd(b, 144)$, so $\gcd(b, 144) = \gcd(c, 144)$. Together, these two imply that $\gcd(a, 144) = \gcd(c, 144)$, and so $a\rho c$.

MARKING:
1 mark for a valid proof of reflexivity
1 mark for a valid proof of symmetry
1 mark for a valid proof of transitivity

(b) **(2 marks)** Determine the equivalence classes. How many equivalence classes are there? Justify your answer.

For each positive divisor d of 144, there is an equivalence class of the form
$$[d] = \{a \in \mathbb{Z}_{\geq 1} \mid \gcd(a, 144) = d\}.$$

As $144 = 2^4 \times 3^2$, there are $(4 + 1) \times (2 + 1) = 15$ distinct positive divisors of 144. This means that there can be at most 15 equivalence classes.

Furthermore, for each positive divisor a of 144, $\gcd(a, 144) = a$. So if a and b are distinct positive divisors of 144, $a \not\rho\, b$, and so they are in different equivalence classes. Therefore there are at least 15 equivalence classes (and hence exactly 15).

MARKING:
1 mark for correctly describing the equivalence classes.
1 mark for identifying that there are 15 equivalence classes with some justification

Note for this question, stating that there is an equivalence class for each positive divisor and that there are $(4 + 1) \times (2 + 1) = 15$ positive divisors of 144 is sufficient justification.

(2) **(3 marks)** Let S be the set of binary strings of length 6. The weight of a binary string is the number of ones it contains, for example weight(110110) = 4. Define a relation ρ on S by

$$s \rho t \qquad \text{if and only if} \qquad \text{weight}(s) \leq \text{weight}(t).$$

Is ρ a partial order? Justify your answer.

The relation is not antisymmetric. Let $s = 100000$ and $t = 000001$. Then $\text{weight}(s) = 1 = \text{weight}(t)$. Thus, $s \rho t$ and $t \rho s$ but $s \neq t$.

Since it is not antisymmetric, the relation is not a partial order.

MARKING:

1 mark for identifying that it is not a partial order
2 marks for providing a valid counter-example

(3) **(2 marks)** Determine whether $\mathbb{Z}_{10} - \{0\}$ is a group under multiplication.

This is not a group. Consider the elements 2 and 5 in $\mathbb{Z}_{10} - \{0\}$. Under the operation of multiplication,

$$2 \times 5 \equiv 10 \equiv 0 \pmod{10},$$

which is not in $\mathbb{Z}_{10} - \{0\}$. Thus this is not closed, and therefore not a group.

MARKING:

1 mark for identifying that it is not a group
1 marks for providing a valid counter-example